

IST 686 Homework 2

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1. Flip a fair coin nine times and write down the number of heads obtained. Now repeat this process 100,000 times. Obviously you don't want to have to do that by hand, so create the necessary lines of R code to do it for you. Hint: You will need both the `rbinom()` function and the `table()` function. Write down the results and explain in your own words what they mean.

```
table(rbinom(n=100000, size=1, prob=0.5))
```

```
##  
##      0      1  
## 50130 49870
```

Since this is bernoulli trial with either success or fail we can consider the 0 to be tails or fail and 1 to be heads or success, in which case we would say that out of 100,000 coin tosses we would have 49,872 heads.

2. Using the output from Exercise 1, summarize the results of your 100,000 trials of nine flips each in a bar plot using the appropriate commands in R. Convert the results to probabilities and represent that in a bar plot as well. Write a brief interpretive analysis that describes what each of these bar plots signifies and how the two bar plots are related. Make sure to comment on the shape of each bar plot and why you believe that the bar plot has taken that shape. Also make sure to say something about the center of the bar plot and why it is where it is.

```
barplot(table(rbinom(n=100000, size=1, prob=0.5)), main="Raw Coin Toss Counts")
```

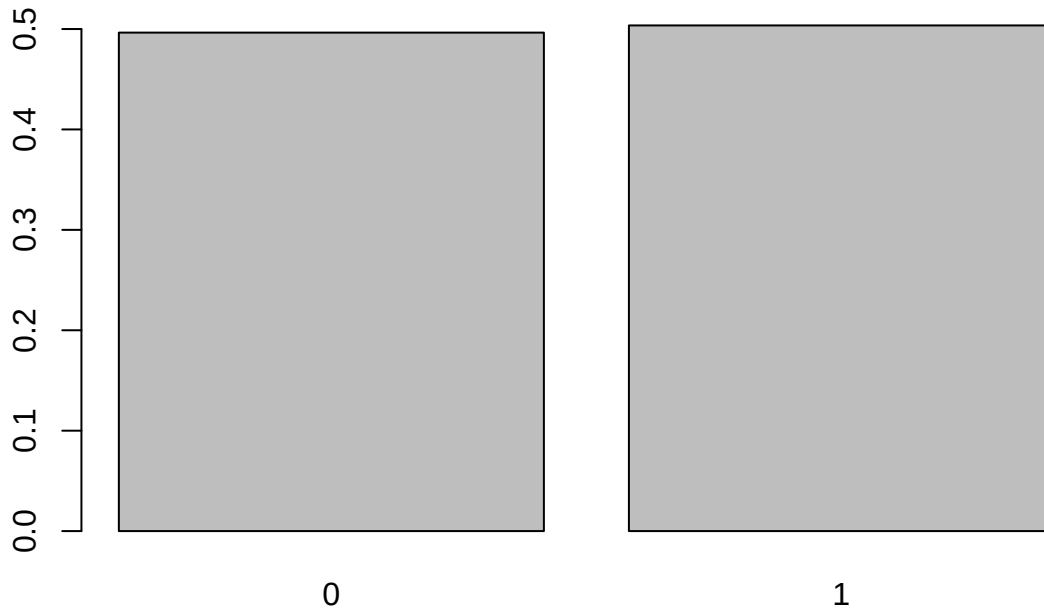


```
prob.table<-table(rbinom(n=100000, size=1, prob=0.5))/100000
prob.table
```

```
##
##      0      1
## 0.49642 0.50358
```

```
barplot(prob.table, main="Barplot of Coin Toss Probabilities")
```

Barplot of Coin Toss Probabilities



The barplot of the raw coin toss counts displays the number of heads and the number of tails that occurred during the 100,000 tosses of the coin. Since the count is fairly close the barplot does not show us much of a difference between the two counts. Likewise, the barplot of the probabilities shows us what the probability of getting heads or tails will be if we toss a coin 100,000 times, which as shown in the output of the probability table and the barplot is 50.06% tails and 49.94% for heads. The information we can gain from looking at the shape of the barplot is that the probabilities are almost perfectly evenly distributed 50/50. If I recall correctly this has to do with the law of large numbers and CLT, as if we only flipped the coin 9 times we would end up with a variety of measurements, sometimes 30/70, 60/40, 40/60, so on and so forth. But as we increase the number of flips we get closer to the true distribution which would be a 50/50 split on a perfectly balanced coin. This barplot compared to a variety of 9 flip barplots would show this visually.

*I want to point out that I am aware that I only have two bars on my barplot and this is due to the `size=1` within the `rbinom()` command. I felt that this was most appropriate given the circumstance but if it was supposed to have multiple bins and show the convergence to normal then I apologize in advance. Well, this is the end of the answer so I guess its not really in advance...

6. One hundred students took a statistics test. Fifty of them are high school students and 50 are college students. Eighty students passed and 20 students failed. You now have enough information to create a two-by-two contingency table with all of the marginal totals specified (although the four main cells of the table are still blank). Draw that table and write in the marginal totals. I'm now going to give you one additional piece of information that will fill in one of the four blank cells: only three college students failed the test. With that additional information in place, you should now be able to fill in the remaining cells of the two-by-two table. Comment on why that one additional piece of information was all you needed in order to figure out all four of the table's main cells. Finally, create a second copy of the complete table, replacing the counts of students with probabilities. What is the pass rate for high school students? In other words, if one focuses only on high school students, what is the probability that a student will pass the test?

```
stu.test<-matrix(c(17,33,3,47), ncol=2, byrow=TRUE)
colnames(stu.test)<-c("Fail", "Pass")
rownames(stu.test)<-c("High School", "College")
stu.test<-as.table(stu.test)
stu.test
```

```
##           Fail Pass
## High School  17   33
## College      3   47
```

I split the code here instead of running it altogether as displayed in the book just beacuse I felt i

```
margin.table(stu.test)
```

```
## [1] 100
```

```
margin.table(stu.test,1)
```

```
## High School    College
##           50           50
```

```
margin.table(stu.test,2)
```

```
## Fail Pass
##    20   80
```

```
stu.test.probs<-stu.test/margin.table(stu.test)
stu.test.probs
```

```
##           Fail Pass
## High School 0.17 0.33
## College     0.03 0.47
```

The reason that one piece of information that was given was all that was needed to finish the contingency table was due to the fact that we already knew that the split was 50/50 between college students and high school students. We also knew that there were a total of 80 passes and 20 fails. This means that if we only had 3 college students fail then the other 17 would have to be high school students. From there we can fill out the table the rest of the way with relative ease.

If we wanted to focus only on the values for high school students we could normalize the probabilities in the contingency table:

17/50

```
## [1] 0.34
```

33/50

```
## [1] 0.66
```

So if we focus only on High School students the probability that a student taking the test will pass is 66%.

7. In a typical year, 71 out of 100,000 homes in the United Kingdom is repossessed by the bank because of mortgage default (the owners did not pay their mortgage for many months). Barclays Bank has developed a screening test that they want to use to predict whether a mortgagee will default. The bank spends a year collecting test data: 93,935 households pass the test and 6,065 households fail the test. Interestingly, 5,996 of those who failed the test were actually households that were doing fine on their mortgage (i.e., they were not defaulting and did not get repossessed). Construct a complete contingency table from this information. Hint: The 5,996 is the only number that goes in a cell; the other numbers are marginal totals. What percentage of customers both pass the test and do not have their homes repossessed?

```
bank.test<-matrix(c(5996,93933,71,0), ncol=2, byrow=TRUE)
colnames(bank.test)<-c("Fail", "Pass")
rownames(bank.test)<-c("Not Default Risk", "Default Risk")
bank.test<-as.table(bank.test)
bank.test
```

```
##              Fail  Pass
## Not Default Risk 5996 93933
## Default Risk      71    0
```

```
margin.table(bank.test)
```

```
## [1] 1e+05
```

```
margin.table(bank.test,1)
```

```
## Not Default Risk      Default Risk
##              99929              71
```

```
margin.table(bank.test,2)
```

```
## Fail  Pass
## 6067 93933
```

```
bank.test.probs<-bank.test/margin.table(bank.test)
bank.test.probs
```

```
##              Fail    Pass
## Not Default Risk 0.05996 0.93933
## Default Risk     0.00071 0.00000
```

```
93933/99929
```

```
## [1] 0.9399974
```

When looking at the number of homeowners who made their payments and were not a default risk AND passed the test we can see that there was a 94% pass rate. (93.99974)